

Snooker Dataset - Exploratory Data Analysis

Axorvian Internship Program | Task 1
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1. Dataset Overview

The dataset used for this analysis is the **Snooker Data (1982-2020)**, obtained from Kaggle. It contains comprehensive records of professional snooker matches spanning nearly four decades. The dataset includes match results, player names, tournament stages, scores, and walkover information.

Dataset Source:

<https://www.kaggle.com/datasets/rusiano/snooker-data-19822020>

Attribute	Details
Total Records	193,530 matches
Total Features	13 columns
Time Period	1982 - 2020
File Used	matches_r.csv
Tool Used	Google Colab, Python, Pandas, Matplotlib, Seaborn

2. Analysis Performed

The following steps were carried out during the exploratory data analysis:

Data Loading: The matches_r.csv file was loaded using Pandas in Google Colab.

Shape Inspection: Dataset contains 193,530 rows and 13 columns.

Data Types: Columns include int64 (tournament_id, match_id, best_of, score1, score2), object (date, stage, player names, URLs, frames_scores), and bool (is_walkover).

Missing Values: Two columns had significant missing values: date (138,711 missing) and frames_scores (123,278 missing). All other columns were complete.

Winner Column: A new 'winner_name' column was derived by comparing score1 and score2 to identify the match winner.

Statistical Summary: Descriptive statistics were computed for all numerical columns including mean, std, min, and max values.

3. Visualizations

Visualization 1: Top 10 Snooker Players by Matches Won (Bar Chart)

A horizontal bar chart was created to display the top 10 players with the most match wins from 1982 to 2020. The chart was built using Seaborn with a viridis color palette.

Visualization 2: Distribution of Match Stages (Pie Chart)

A pie chart was created to show the distribution of matches across different tournament stages such as Last 128, Last 64, Last 32, Quarter-final, Semi-final, and Final. This helps understand at which stage the majority of matches are played.

4. Key Findings

1. Total Matches Analyzed: 193,530 professional snooker matches from 1982 to 2020.
2. Top Player: John Higgins leads with 1,106 match wins, followed by Ronnie O'Sullivan and Mark Williams.
3. Most Common Stage: Last 128 accounts for 19.9% of all matches, indicating most tournaments have large early-round fields.
4. Average Winner Score: The average winning score is 4.21 frames per match.
5. Walkover Matches: 4,596 out of 193,530 matches (approximately 2.4%) were walkovers, meaning one player did not compete.
6. Missing Dates: Over 71% of date entries are missing, suggesting older records were not digitized with complete date information.

5. Conclusion

This exploratory data analysis of the Snooker dataset (1982-2020) provided meaningful insights into professional snooker over nearly four decades. John Higgins emerged as the most dominant player by total wins. The stage distribution highlights how tournaments are structured with large early rounds narrowing down to finals. The dataset is rich in match-level information and can be further explored for advanced analysis such as player head-to-head records, performance trends over time, and tournament-level statistics.